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Keyword Filtering for Digital Content Recommendation

Abstract

Methods, systems, and apparatus, including computer-readable storage media, for keyword list filtering as part of identifying digital content responsive or relevant to a search query or request for content. A user, such as a content provider, may generate a keyword list associated with digital content of the content provider. Keyword lists, however, may be built over the course of years and can grow to include millions of keywords. Further, these keyword lists are often not maintained in line with changes in a content provider's digital content delivery strategy or context. An artificial intelligence (AI) model may be trained to generate a summary of the digital content associated with the content provider. That summary, along with the keyword list of the content provider, is provided as input into the AI model, which is trained to provide, as output, a recommendation to keep or remove a keyword from the keyword list.

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Background/Summary

BACKGROUND

[0001] Digital content delivery systems provide digital content, e.g., text, audio, images, videos, etc., in response to search queries or requests for content. To determine what digital content should be provided in response to a request, the digital content delivery system can maintain a list of keywords associated with the digital content. A negative keyword list is a list of keywords, which, if present in a request for digital content or user input associated with the request, indicates that digital content associated with the list of keywords should not be sent in response to the request. Negative keyword lists may span millions of keywords added over the course of years. Keywords in these lists become outdated over time, which causes some digital content to be excluded that would otherwise be appropriate or relevant to provide in response to a digital content request.

BRIEF SUMMARY

[0002] Aspects of the disclosure for keyword list filtering as part of identifying digital content responsive or relevant to a search query or request for content. A user, such as a content provider, may generate a keyword list associated with digital content of the content provider. Keyword lists, however, may be built over the course of years and can grow to include millions of keywords. Further, these keyword lists are often not maintained in line with changes in a content provider's digital content delivery strategy or context. An artificial intelligence (AI) model may be trained to generate a summary of the digital content associated with the content provider. That summary, along with the keyword list of the content provider, is provided as input into the AI model, which is trained to provide, as output, a recommendation to keep or remove a keyword from the keyword list.

[0003] Aspects of the disclosure also provide for an AI model training process for training a model to recommend keywords more accurately for removal from a keyword list. The training process integrates user feedback over the course of dozens or hundreds of keyword filtering recommendations generated by the model, to tune the model to later process thousands or millions of remaining keywords accurately and without user intervention. The feedback can clarify, or correct recommendations generated by the AI model, and/or natural language explanations that the AI model is also trained to generate. The explanations provide reasons or context as to why the AI model generated a particular recommendation to remove or keep a keyword.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a block diagram of an example keyword filter system implemented as part of a digital content delivery system, according to aspects of the disclosure.

[0005] FIG. 2A is a block diagram of the example keyword filter system in a processing mode, according to aspects of the disclosure.

[0006] FIG. 2B is a block diagram of the example keyword filter system in a training mode, according to aspects of the disclosure.

[0007] FIG. 3A is an example of a user interface for generating and receiving a summary of digital content, according to aspects of the disclosure.

[0008] FIG. 3B is an example of a user interface for modifying the summary of a content provider.

[0009] FIG. 3C is an example of a user interface for managing keyword lists.

[0010] FIG. 3D is an example of a user interface for displaying recommendations and explanations generated by an AI model for keyword filtering, according to aspects of the disclosure.

[0011] FIG. 3E is an example of a user interface displaying or outputting an example recommendation and explanation during training of an AI model for keyword filtering, according to aspects of the disclosure.

[0012] FIG. 3F is another example of a user interface for providing feedback during training of an AI model for keyword filtering, according to aspects of the disclosure.

[0013] FIG. 3G is an example of a user interface after the system trains on a batch of keywords, according to aspects of the disclosure.

[0014] FIG. 4A is a flow diagram of filtering a keyword list using a trained AI model, according to aspects of the disclosure.

[0015] FIG. 4B is a flow diagram of an example process for training an AI model to filter a keyword list, according to aspects of the disclosure.

[0016] FIG. 5 is a block diagram illustrating one or more model architectures, such as for deployment in a data center housing a hardware accelerator on which the deployed models will execute for keyword list filtering, according to aspects of the disclosure.

[0017] FIG. 6 is a block diagram of an example computing environment for implementing the keyword filter system.

DETAILED DESCRIPTION

[0018] Aspects of the disclosure are generally directed to filtering a keyword list that is used to identify digital content responsive to a search query or request for content. A user, such as a content provider, may generate a keyword list associated with digital content. The keyword lists may be processed by one or more artificial intelligence (“AI”) models that are trained to generate recommendations for keywords to remove from these keyword lists.

[0019] An AI model may be trained to generate a summary of the digital content associated with the content provider. The summary can be, for example, a series of statements or propositions related to the content provider, the digital content provided by the content provider, and/or the audience intended for the digital content provided by the content provider. The summary, along with the keyword list of the content provider, may be provided as input into the same or another AI model, which may be trained to provide, as output, a recommendation to keep or remove a keyword from the keyword list. The model can also output an explanation in natural language, explaining reasons for the recommendation to keep or remove a keyword from the keyword list.

[0020] Aspects of the disclosure also provide for a model training process for training a model to recommend keywords more accurately for removal from a keyword list. The training process integrates user feedback over the course of dozens or hundreds of keyword filtering recommendations generated by the model, to tune the model to later process thousands or millions of remaining keywords accurately and without user intervention.

[0021] Keyword lists may be built over the course of years and can grow to include millions of keywords. Further, these keyword lists are often not maintained in line with changes in a content provider's digital content or its digital content delivery strategy. Aspects of the disclosure can provide for at least the following technical advantages. Omitting keywords from the keyword list results in fewer keywords to process, while allowing for more accurate content delivery, overall. Having a keyword list with fewer keywords to process increases the computational efficiency of the system. Fewer keywords also can decrease the processing power and network overhead to communicate those keywords from a storage device to a processing device. A keyword list with fewer keywords requires less memory to store.

[0022] As a digital content delivery system may include digital content and keyword lists from hundreds or thousands of content providers, filtering keywords for even a subset of content providers can have significant effect on the efficiency of the content delivery system in parsing keyword lists and providing content in response to received queries and requests. Additionally, content can be provided in response to queries that would otherwise not have been provided because of one or more outdated keywords, improving content delivery accuracy. However, by filtering the keyword list, processing power and network overhead is decreased as providing replacement content for inaccurate and/or unwanted content would not be necessary, due to the filtered keyword list.

[0023] FIG. 1 is a block diagram of an example keyword filter system **100** implemented as part of a digital content delivery system **101**, according to aspects of the disclosure. The digital content delivery system **101** can receive user requests for content **185** from a user computing device **180**. The user computing device **180** can be one of multiple user computing devices, which may also include user computing devices **180A-N**. The user request **185** can be, for example, a search query to a search engine implemented as part of the digital content delivery system **101**, or other request for content. In some examples, input to the digital content delivery system automatically includes a request for content, for example for providing digital content that may be relevant in response to user activity, even if the input does not include an explicit content request.

[0024] The digital content delivery system **101** can receive the user request **185** for content at a request response engine **150**. The engine **150** accesses a keyword list repository **115** and a digital content repository **175**. The repositories **115** and **175** can be implemented, for example as one or more storage devices stored in one or more physical locations. In some examples, the repositories **115** and **175** form part of the same database or repository stored on the same storage devices.

[0025] The keyword list repository **115** stores various keyword lists associated with different providers of digital content. The keyword list can include negative keywords, which may be present in a search query or otherwise included as part of input on a computing device that is the recipient of digital content. A negative keyword is a keyword in which, if present in a search query or request for content, indicates that digital content associated with the list the negative keyword is a part of should not be provided in response to the query or request. The digital content delivery system **101** can be configured to determine that a query or user input includes one or more listed negative keywords, and not provide digital content associated with the keyword list to the recipient computing device.

[0026] In some examples, the keyword list can be a positive keyword list. A positive keyword is a keyword in which, if present in a search query or request for content, indicates that digital content associated with the list the positive keyword is a part of may be provided, or is more likely to be provided, in response to the query or request. The digital content delivery system **101** can be configured to determine that a query or user input includes one or more listed positive keywords, and provide digital content associated with the keyword list to the recipient computing device.

[0027] The request response engine **150** can determine, for a particular content provider, whether digital content associated with the content provider is to be provided in response to the user request **185**. For example, if the user request **185** accompanies user input for a search query related to “sports,” the engine **150** can determine that a keyword list for a given content provider includes the negative keyword “sports.” In response to the user request **185**, the engine **150** does not provide digital content from the digital content repository **175** associated with the keyword list of the given content provider, at least because the user request **185** contained a negative keyword. If the request response engine **150** determines that there is not a negative keyword in the user request **185**, the engine **150** can provide digital content **190** that is responsive to the user request **185**.

[0028] The engine **150** can provide the digital content **190** based on a variety of different heuristics, rules, rankings, or priorities. Digital content **190** can be, for example, text, images, videos, audio, and so on. The digital content **190** can correspond to content that is predicted to be responsive to the user request **185** or considered to be potentially of interest to a user making the request **185**. The request response engine **150** can be configured to predict digital content responsiveness and/or digital content relevancy to the user, for example using one or more AI models trained to perform these predictions. Example digital content can include advertisements for products or services that are predicted to be relevant to the user making the request **185**, and/or predicted to be responsive to the request **185** itself.

[0029] The keyword filter system **100** can receive a keyword list **105**. The keyword list **105** can be stored on the keyword list repository **115** initially, or, in some examples, provided from another source, e.g., from a computing device associated with a content provider. The keyword filter

system **100** is configured to generate a filtered keyword list **110**. The keyword filter system **100** can receive multiple different keyword lists associated with the same or different content providers and filter the keywords according to aspects of the disclosure. Filtering a keyword list can refer to the system generating recommendations to keep or remove keywords from the keyword list **105**. The keyword list **105** can be a negative keyword list or a positive keyword list, according to aspects of the disclosure. In some examples, the keyword list **105** may include both positive and negative keywords, indicated by some additional data, tag, field, etc.

[0030] The filtered keyword list **110** can include a subset of keywords that the keyword filter system **100** recommends keeping. The subset of keywords that the keyword filter system **100** recommends keeping may include fewer keywords than the initial keyword list **105**. The filtered keyword list **110** can be stored in the repository **115** and accessed by the request response engine **150** for determining what digital content to provide in response to a request.

[0031] As an example, a keyword list may relate to digital content from a content provider relating to baseball. The keyword list may include terms from sports unrelated to baseball, e.g., negative keywords “basketball,” “soccer,” or “tennis.” In some examples, the keyword list may include these unrelated terms to prevent digital content relating to baseball from being sent to computing devices in response to queries or input containing these keywords. However, over time, the digital content from the content provider may go from being more specific, e.g., limited to only baseball, to being about sports in general.

[0032] The shift in scope from a specific sport, e.g., baseball, to sports in general can be reflected in a summary generated by the keyword filter system **100**. The summary can be, for example, a series of statements or propositions related to the content provider, the digital content provided by the content provider, and/or the nature of the intended audience for the digital content provided by the content provider. For example, the summary may include statements describing the category of digital content provided by the content provider. The category of digital content provided by the content provider may include, for example, informative content, entertainment, satirical content, advertisements, etc. The summary may also indicate the topics covered by the digital content. The topics of categories may include, for example, sports, news, products, services, technology, etc. The summary may also include statements related to the intended audience for the content provider. The intended audience may be, for examples, individuals, organizations, residents of a certain geographic region, consumers of content in a particular spoken or written language, etc. Techniques for generating the summary are described with reference to FIG. 2A-2B, herein.

[0033] In the “sports” example described above and herein, the keyword filter system **100** can receive an input keyword list and a summary, and provide recommendations to remove specific sports, e.g., “basketball,” “soccer,” or “tennis,” from the keyword list. An example output can be a recommendation to remove the negative keyword “basketball” from the keyword list, with a provided explanation being that “basketball is a type of sport and therefore relevant to sports content in general.”

[0034] The filtered keyword list **110** can require less storage space to store in the keyword list repository **115**. Overall, the digital content delivery system **101** can perform more efficiently, as less data is required to transmit between the repository **115** and the request response engine **150**. In some examples, if the request response engine **150** processes input keyword lists using a technique that executes in linear time, e.g., in time proportional to the length of the input keyword list, the engine **150** can process the filtered keyword list **110** in fewer processing cycles versus an unfiltered keyword list. The engine **150** using the filtered keyword list **110** to determine whether a negative keyword is in the user request **185** can also result in digital content not being omitted that would otherwise be responsive or relevant to the request.

[0035] Aspects of the disclosure can be implemented as a software application, e.g., a web application through a browser, a mobile application, a desktop application, etc., configured to manage keyword lists corresponding to different user accounts. The keyword filter system **100**

described herein may be a feature of a computing platform, e.g., the digital content delivery system **101**, that can be accessed through the application. In some examples, the keyword filter system **100** may be enabled for periodic keyword list filtering, e.g., on a quarterly basis or in response to the digital content delivery system **101** receiving input to add keywords to an existing list, or to add a new keyword list altogether. In some examples, the keyword filter system **100** may automatically perform the processes described herein, generating a recommended output list, which may be used to update existing keyword lists.

[0036] FIG. 2A is a block diagram of the example keyword filter system **100** in a processing mode, according to aspects of the disclosure. The keyword filter system **100** can receive a keyword filter request **205** from a user computing device **250**. The user computing device can be associated with a content provider, for whom digital content is sent through the digital content delivery system **101**. Input and output can be received and sent between the keyword filter system **100** and the computing device **250** through a user interface **300**. The user interface **300** can include user interfaces **300A-300G**, described herein and with reference to FIGS. 3A-3G. The user interface **300** can be, for example, an API, a web page, a standalone computer program, etc., which can be at least partially implemented in the user computing device **250** and/or on devices implementing the keyword filter system **100** or the content delivery system **101**.

[0037] The keyword filter system **100**, in response to the keyword filter request **205**, can receive and process one or more negative keyword lists through artificial intelligence (AI) model **200**. AI model **200** is one of multiple AI models that may be implemented, including AI models **200A-200N**. In some examples, all the processing described herein with reference to an AI model is performed by the AI model **200**. In some examples, processing described as being performed by an AI model is performed across multiple different AI models **200A-200N**. In some examples, the AI model **200** may be a machine learning model, e.g., a large language model or large generative model.

[0038] The AI model **200** can receive, as input, keyword lists corresponding to the content provider associated with the keyword filter request **205**. The received lists can be preloaded onto the keyword list repository **115** of the digital content delivery system **101** and selected from a larger selection of lists maintained by the repository **115** and associated with the content provider. The system can be configured in some examples to aggregate multiple input lists and generate a composite input list of unique keywords across the multiple lists. The keyword filter system **100** can receive multiple lists that may or may not overlap with one another, e.g., having the same or similar keywords.

[0039] The AI model **200** can receive a summary of the content provider, in addition to the input keyword lists for filtering. The AI model **200** may be trained to generate a summary of the digital content associated with the content provider. The summary can be in natural language, summarizing aspects of the digital content associated or provided by the content provider, digital content provided by the content provider, and/or the audience intended for the digital content. As part of generating the summary, the AI model **200** is configured to incorporate any additional context or information about the digital content from other sources, for example, based training on a larger corpus of data, which may include the digital content or information related to the digital content.

[0040] The AI model **200** can generate the summary from digital content **215** associated with the content provider. The digital content **215** may or may not include digital content intended for delivery in response to search queries or content requests. For example, the keyword filter system **100** can receive a homepage for a content provider's website or other web page or web content, or other videos, images, and/or text associated with the content provider and/or the digital content typically created or provided by the content provider. In some examples, the digital content **215** includes digital content intended for delivery in response to a digital content request at the digital content delivery system **101**.

[0041] The AI model **200** can generate the summary from the digital content **215** according to any one of a variety of different natural language processing techniques for summarizing media, such as text. For example, the AI model **200** can be a deep neural network trained on training examples of text and summaries of the text. In some examples, the AI model **200** can implement a seq2seq framework or other type of encoder-decoder framework, to convert input text into an output summary. In some examples, if the AI model **200** is trained to receive input in modalities other than text, the AI model **200** can also be trained to convert non-text input into a text representation, e.g., transcribing audio, detecting text in input video or images, etc. The AI model **200**, in some examples, can apply an attention mechanism as part of an encoder-decoder framework, to generate the summary.

[0042] In some examples, the digital content **215** received by the keyword filter system **100** can include information specific to digital content for a particular input list. For example, the digital content delivery system **101** may maintain multiple lists for a content provider, each list corresponding to a type of digital content provided by the user. Depending on the input lists, the system can receive information describing or characterizing the digital content associated with a list and provide that as part of the input to the model for generating a summary.

[0043] In some examples, the keyword filter system **100** may retrieve a pre-generated summary associated with a content provider. In some examples, the keyword filter system **100** may receive a summary as input, e.g., as part of the request **205**, instead of generating the summary. The keyword filter system **100** can be configured to receive, e.g., through the user interface **300**, user input for editing or revising the generated summary.

[0044] The keyword filter system **100** can process an input list of keywords and the summary through the AI model **200** trained to filter the keywords of the input list. For each keyword, the AI model **200** can generate a recommendation to remove or keep the keyword in the input keyword list. The AI model **200** can be trained to generate the recommendation as a type of classification task for the input keyword list and summary. The classification task can define two labels, e.g., “keep” and “remove.”

[0045] The recommendation generated by the AI model **200** can be done based on, for example, relevance of the keyword to the input summary. One measure of relevance can include typological or lexical similarity, e.g., how similar are keywords to words used in the content provider summary. In some examples, a measure of relevance can be based on semantic similarity, e.g., how similar in meaning are the keywords to the words used in the user summary. For example, the AI model **200** is more likely to have the same recommendation for two keywords that are etymologically distinct but semantically similar. For example, in filtering a keyword list for a content provider that has shifted from providing digital content specifically for basketball, to content about sports in general, the AI model **200** can recommend both that the negative keyword “soccer” and the negative keyword “football” be removed from the filter list, as the terms can refer to the same game in different parts of the world. The AI model **200** can also identify similar keywords written in different languages.

[0046] The recommendation to keep or remove a given keyword can be based on a score generated by the AI model **200**. Higher values of the score can correlate to a higher chance of recommendation to keep the keyword in an input list. Lower values of the score can correlate to a higher chance of recommendation to remove the negative keyword from the input list. In some examples, The AI model **200** can be trained to generate a score that is converted into an output classification, e.g., “keep” or “remove” the keyword. For example, if the score meets or exceeds a predetermined threshold, the AI model **200** can be trained to output a recommendation to keep the negative keyword. If the score meets or falls below another predetermined threshold, the AI model **200** can be trained to output a recommendation to remove the negative keyword from the input list.

[0047] The AI model **200** can also generate, along with a recommendation to keep or remove a keyword from a list, an explanation for the recommendation. The explanation can be expressed, for

example, in natural language, as text, although the explanation in some examples may be generated in the form of audio, images, video, etc. The explanation can provide one or more statements for the recommendation, based on a measure of relevance used to train the AI model **200**.

[0048] For example, the explanation can relate to the semantic similarity of the keyword with other keywords, or text in the content provider summary. If the content provider is summarized as “shifting from providing content related to basketball, to providing content about sports in general,” then an explanation for the recommendation to remove “soccer” as a negative keyword can be “soccer is a type of sport, and the content provider provides sports content, generally.”

[0049] As another example, an explanation can relate to other measures of relevancy, such as lexical relevance. In the sports content provider example, if the summary indicates that the content provider provides content related to “soccer,” then the explanation as to why the AI model **200** recommends removing the negative keyword “football” can be “football and soccer can refer to the same sport.”

[0050] As described herein, the AI model **200** can be trained according to a training process that integrates feedback data, which can be related to a recommendation generated by the AI model **200**, an explanation generated by the AI model **200**, or both. In some examples in which the AI model **200** integrates feedback data, the feedback data can form at least a partial basis for the explanation. In the sports content provider example, if the AI model **200** previously recommended keeping the negative keyword “chess” with the explanation that “chess is not a sport,” the AI model **200** may later receive feedback data indicating that chess is considered a sport by the content provider. Thereafter, the AI model **200** may generate a recommendation to “keep” the negative keyword “chess,” with the explanation that “chess is considered a sport.”

[0051] A filtered keyword list **210** is an example of an output list generated by the AI model **200**. The filtered keyword list **210** may only include keywords that remain after the system filters the keywords of the input list. In some examples, the AI model **200** may generate, as an output list, a change log or other data representing recommended changes to the input keyword lists processed as part of responding to the request **205**. The keyword filter system **100** can receive the change log as input for making the changes to the input keyword list. The change log may be reviewed or modified, for example by the content provider, before the keyword filter system **100** receives the change log as input. In some examples, the filtered keyword list **210** can include all the negative keywords from the input list, with an added field storing the recommendation of the AI model **200**.

[0052] In some examples, the filtered keyword list **210** can also include the explanation generated by the AI model **200** corresponding to the recommendation to keep or remove a given keyword, e.g., a negative or positive keyword. In some examples, the keyword filter system can also include as part of the output list any identifiers associated with the keyword, e.g., a name or other identifier from the input list or lists the keyword is a part of, and/or identifiers corresponding to the digital content that is not served in response to search queries including the keyword.

[0053] In some examples, the AI model **200** can be trained to recommend additional negative keywords to include in a keyword list. The added keywords may be suggested, for example, based on a predicted relevance the keyword may have to the content provider, digital content of the content provider, or the intended audience of the digital content. The AI model **200** can at least partially determine the relevance of a negative keyword to add to the keyword list, for example based on its relevance to text provided as part of the input summary for the content provider. As an example, if the summary indicates that the content provider is a sports content provider, but does not provide content related to “tennis,” the AI model **200** can be trained to identify the context in which the term “tennis” is used in the summary, to propose that “tennis” also appear as a negative keyword.

[0054] In some examples, the AI model **200** can be trained to generate new positive keywords, based on an input positive keyword list and a summary of a content provider, and recommend the inclusion of those positive keywords in the keyword list. In some examples, different AI models are

implemented for processing keyword lists with negative and positive keywords, separately. In some examples, the AI model **200** is trained to process both keyword lists with both negative and positive keywords, as described herein.

[0055] Filtering keyword lists using one or more AI models can increase computational efficiency. For example, by omitting keywords from a keyword list based on recommendations of the AI model **200**, the subsequent scanning of the lists to determine digital content delivery is more computationally efficient. The reduced bandwidth, processing, and memory usage scale on a platform managing many keyword lists, each with potentially millions of keywords, across different content providers and various categories of digital content. Filtered keyword lists make identifying responsive digital content more accurate, and therefore, reduces the need to have to identify alternative content in response to a second request from a user after the original digital content provided was non-responsive. Preventing subsequent request for contents increases efficiency by reducing processing power consumption and network overhead.

[0056] FIG. 2B is a block diagram of the example keyword filter system **100** in a training mode, according to aspects of the disclosure. The keyword filter system **100** can include a training engine **225** configured for fine-tuning or at least partially re-training the AI model **200** in response to feedback data **255**. Feedback data **255** is received through the user interface **300**, for example in response to feedback or corrections to candidate recommendations and/or explanations generated by the AI model **200**. Examples of candidate recommendations and/or explanations and of the feedback data **255** are provided in FIGS. 3A-3G and the corresponding description.

[0057] The AI model **200** may be pre-trained, for example on a corpus of text or other training data to perform any of a variety of different natural language processing tasks, such as text recognition, translation, text generation, etc. Training as described herein, e.g., with reference to FIG. 2B, can refer to either fine-tuning a pre-trained model, and/or retraining at least portions of the pre-trained model, e.g., by adding, removing, or modifying weight and/or bias parameter values.

[0058] During training, the keyword filter system **100** may receive or generate batches of keywords from input lists, e.g., from the user computing device **250** and/or the repository **115**. The batches can be of various sizes, e.g., batches of 10-20 keywords per batch, although in some examples the AI model **200** is configured to process the negative keywords one at a time. For each keyword in the batch, the model outputs a recommendation to keep or remove the keyword from the keyword list. The AI model **200** also outputs an explanation for the recommendation.

[0059] At the end of processing the batch, the keyword filter system **100** can receive the model output, e.g., a recommendation and explanation, for each keyword in the batch, and provide the output to the user computing device **250** through the user interface **300**. The user interface **300** is configured to receive feedback data **255** on the recommendations and/or explanations provided for the batch of keywords. The feedback data **255** can be, for example, a user rejection or acceptance of the recommendation for a given keyword. The feedback data **255** can also include revisions to the explanations provided by the AI model **200** for the recommendation to keep or remove a keyword from the list.

[0060] After receiving the feedback data **255**, the training engine **225** can fine-tune or at least partially retrain the AI model **200** to incorporate any feedback from the previous batch and process a new batch of keywords. For example, the feedback data **255** can form a new or updated label for the input keyword, corresponding to a new or updated recommendation or explanation. The training engine **225** can compute a loss function between the updated label and the current recommendation and/or explanation and use the loss to update weight or bias parameter values of the AI model **200**, e.g., using backpropagation and gradient descent with model parameter updates. The training engine **225** can implement any of a variety of different techniques for fine-tuning a large language model with supervised learning, in examples in which the AI model **200** is a large language model.

[0061] The keyword filter system **100** can receive the model output corresponding to another batch

of keywords, and receive additional feedback data, if any. The keyword filter system **100** can also process the remaining keywords in the list using the AI model **200** updated with any feedback data provided during earlier training iterations. The AI model **200** can be updated with updated model parameter values **230**, generated by the training engine **225**.

[0062] Fine-tuning the AI model **200** as described herein allows for the AI model **200** to be efficiently adapted for accurate keyword list recommendations, on an individual content provider basis. For a digital content delivery system, which may manage different content providers with respective keyword lists and digital content, aspects of the disclosure provide for accurate keyword list recommendations. Dozens or hundreds of keywords may be processed in batches for training the AI model **200** to recommend thousands or tens of thousands of keywords automatically and accurately for filtering, overall.

[0063] FIGS. **3A-3G** show examples of user interfaces that may be implemented as part of the user interface **300** for keyword list filtering, content summary generation, and/or keyword list filter training, as described herein. User interfaces described herein, including, for example user interfaces **300A-300G**, can be implemented in any of a variety of different locations and according to any combination of hardware, firmware, and/or software. A user interface may be implemented at least partially on a user computing device, e.g., as a web page loaded on an internet browser, or as a computer program.

[0064] Input or output provided through a user interface can be provided directly to the keyword filter system **100**. In some examples, input or output may be directed to a separate device or service configured for at least partially implementing the user interface, which then redirects traffic to the keyword filter system **100** to execute operations in response to the input. The keyword filter system **100** may provide output directly to a computing device at least partially implementing a user interface, or, in some examples, direct traffic first to the separate device or server.

[0065] An element of a user interface can provide information according to one or more modalities, e.g., text, audio, video, images, etc. An element may be interacted with, e.g., using a user input such as a mouse click, keyword input, touchscreen tap, etc., An element configured to receive input can be, for example, a button, a text field, a drop-down list, etc. The interface can receive input and a system, e.g., the keyword filter system **100**, can be configured to perform one or more operations in response to the input. The keyword filter system **100** can generate output, which can be sent to a computing device through a user interface, for displaying or otherwise outputting data, e.g., as text, audio, images, video, etc. In FIGS. **3A-3G**, text in quotation marks appearing in an element of a user interface is an example of text that may appear as part of the element or is an example of text that may be part of input to the element.

[0066] FIG. **3A** is an example of a user interface **300A** for generating and receiving a summary of digital content, according to aspects of the disclosure. The user interface **300A** can include a digital content summary element **310A** and a summary feedback element **320A**. The digital content summary element **310A** can display or output a summary of digital content received from a content provider. In the user interface **300A**, the element **310A** displays, as an example, “Content provider is an organization that offers informative and entertaining content covering a wide range of sports.” The summary feedback element **320A** can be a user-interactable element for receiving feedback for the summary provided in the element **310A**. In FIG. **3A**, the summary feedback element **320A** is blank.

[0067] FIG. **3B** is an example of a user interface **300B** for modifying the summary of digital content. The digital content summary element **310A** can receive input for modifying the summary after being generated by the keyword filter system **100**. In some examples, the keyword filter system **100** proceeds as described herein, without user input to modify the generated summary.

[0068] As shown in the user interface **300B**, the summary in the digital content summary element **310A** is modified to include an addition: “The content provider covers sports news, in particular French sports news.” The addition provided to the summary modifies the summary to expand in

scope or place particular emphasis on an aspect of the summary. In the example in FIG. 3B, the summary is expanded to include news, and places an emphasis on French sports news, in particular. [0069] The summary feedback element **320A** can receive feedback or modifications to the summary, for example, to limit the scope of the summary or to provide additional clarifications. In the example in FIG. 3B, the summary feedback element **320A** receives the statement: “The content provider does not provide coverage for eSports.” This statement can be integrated as feedback, such that the keyword filter system **100** receiving the summary as input for keyword filtering explicitly does not associate eSports, e.g., video games or virtual gaming, as an example of a sport indicated in the summary. One reason for the feedback can be to clarify or specifically limit the scope of how some terms in a summary are interpreted by the keyword filter system **100**. Whereas a broader interpretation of the phrase a “wide range of sports” may suggest also including eSports, the feedback provided to the element **320A** explicitly causes the keyword filter system **100** to not associate “eSports” with “sports.”

[0070] FIG. 3C is an example of a user interface **300C** for managing keyword lists. The user interface **300C** includes a list selection element **305C** for selecting one or more keyword lists for processing or training, and elements **310C-335C** displaying or outputting various information related to the selected keyword lists.

[0071] Elements **310C**, **315C**, and **320C** can provide information about all the keyword lists for a given content provider. Total negative keywords element **310C** can display or output information related to the total number of negative keywords across all the keyword lists associated with the content provider. Element **310C** shows, as an example, that there are 822 negative keywords in total. Total unique keywords element **315C** can display or output information related to the total number of unique negative keywords across all the keyword lists. Because different keyword lists can overlap on multiple keywords, the element **315C** can provide insight into the number of keywords that appear across the keyword lists, without counting multiple keywords multiple times. The element **315C** shows, as an example, that there are 445 unique keywords across all the keyword lists. Total keyword lists element **320C** can display or output information related to the total number of keyword lists associated with the content provider. The element **320C** shows, as an example, that there are 12 keyword lists overall associated with the content provider.

[0072] The elements **325C**, **330C**, and **335C** provide information similar to the elements **310C**, **315C**, and **320C**, but only for the keyword lists selected to be input to the keyword filter system **100**. For example, total selected keyword lists element **335C** shows that seven keyword lists are selected. Of these seven keyword lists, total selected unique keywords element **330C** shows that there are 235 unique keywords, and total selected negative keywords element **325C** shows that there are 538 keywords overall.

[0073] FIG. 3D is an example of a user interface **300D** for displaying recommendations and explanations generated by an AI model for keyword filtering, according to aspects of the disclosure. Keep list element **305D** and remove list element **306D** can display or output information related to candidate keywords that the AI model recommends keeping or removing from an input keyword list. A candidate keyword is a keyword that has an initial recommendation for keeping or removing but is generated as part of training, and still awaiting user confirmation or feedback. The keep list element **305D** can display or output the number of remaining candidate keywords to keep, shown by the example text “Candidates to keep (1)” in FIG. 3D. The remove list element can display or output the number of remaining candidates to remove, shown by the example text “Candidates to remove (10).”

[0074] Keyword display element **310D** and keyword display element **311D** can display or output the next candidate keyword to review for keeping or removing, respectively. The recommendation element **315D** and recommendation element **316D** can display or output the recommendation for the keywords displayed or output from the elements **310D** and **311D**. Explanation element **320D** and explanation element **321D** can display or output the explanation corresponding to the

recommendations displayed or output from the elements **315D** and **316D**.

[0075] Accept recommendation element **325D** and accept recommendation element **326D** are configured to accept user input for confirming that the recommendations and explanations for the keywords displayed in elements **310D** and **311D** are correct. Reject recommendation element **330D** and rejection recommendation **331D** are configured to receive input indicating that either one or both recommendations and explanations displayed, or output is incorrect. The keyword filter system **100** upon receiving input through reject recommendation elements **330D** and **331D** can advance to user interface **300F** as described herein for receiving feedback data.

[0076] FIG. **3E** is an example of a user interface **300E** displaying or outputting an example recommendation and explanation during training of an AI model for keyword filtering, according to aspects of the disclosure. Keyword display element **310E** displays the current keyword “Montreal.” Recommendation display element **315E** displays the current recommendation to remove the keyword from the keyword list. Explanation display element **320E** displays the explanation corresponding to the recommendation to reject the keyword “Montreal”: “Content Provider interacts with French-speaking Canadians, and Montreal is a French-speaking city in Canada.”

[0077] In this example, the content provider may, at some point after the creation of its keyword list, choose to specifically focus on digital content requests from French speakers from France, instead of French-speaking people, generally. However, over time, the content provider's strategy or focus shifted to include French speakers in other parts of the world, such as Montreal. The keyword filter system is configured to identify this shift, for example from the summary generated for the example content provider.

[0078] In processing the keyword “Montreal,” the keyword filter system is configured to recommend removing the keyword, at least because the summary indirectly or directly indicates that the content provider interacts with French speakers around the world. As a potential source for the explanation and recommendation to reject the negative keyword “Montreal,” the keyword filter system can determine from the summary that digital content associated with the content provider is provided in French, a language that is highly associated with Montreal and other parts of the French-speaking world. The keyword filter system's recommendation can reduce the total keyword list size and prevent potentially relevant digital content from not being sent in response to requests that otherwise would not have received the digital content because of an outdated negative keyword.

[0079] FIG. **3F** is another example of a user interface **300F** for providing feedback during training of an AI model for keyword filtering, according to aspects of the disclosure. Keyword display element **310F** displays negative keyword “Chess,” recommendation display element **315F** displays the recommendation from the AI model to “Keep” the negative keyword “Chess” in the keyword list. Explanation display element **320F** initially provides an explanation for keeping the negative keyword as the text not in bold and underline: “Chess is not a sport and content providers provide informative and entertaining content related to a wide variety of sports.” Referring to the example in FIG. **3A** and the summary display element **310A**, content provider may be “Content provider is an organization that offers informative and entertaining content covering a wide range of sports.” Based on the summary, the keyword filter system **100** recommends keeping the negative keyword “Chess,” with the explanation that chess is not a sport and therefore would not be covered by the content provider.

[0080] In this example, however, the content provider may also consider chess to be a sport, and therefore may reject the recommendation to keep “Chess” as a negative keyword on the keyword list. The user interface **300F** is configured to receive feedback data, for example in the form of a clarification or correction in the recommendation display element **320F**. In this example, the recommendation display element **320F** receives the bolded and underlined feedback data shown in FIG. **3F**: “Chess is considered at least by some to be a sport and content provider also provides content related to chess.” The user interface **330F** can receive input, for example from the reject

recommendation element **330E**, to cause the rejection of the recommendation and the modified explanation to be sent to the keyword filter system **100** as feedback data. The keyword filter system **100** can use the feedback data as described herein to fine-tune the AI model **200** for generating the keyword recommendations and use the updated AI model **200** for future recommendations.

[0081] FIG. **3G** is an example of a user interface **300G** after the keyword filter system **100** trains on a batch of keywords, according to aspects of the disclosure. The user interface **300G** can include an accuracy element **305G**, indicating a percentage or some other metric for displaying or outputting the accuracy of the system during training. In the user interface **300G**, the element **305G** reports 93% accuracy from the last training batch of keywords. One measure of accuracy can be the number of recommendations and/or explanations that did not receive feedback over the total number of recommendations and/or explanations generated for the last training batch.

[0082] Batch completion element **310G** can display or output the number of keywords in which a recommendation was generated, and feedback was requested. In the user interface **300G**, the batch completion element **310G** reports that fifteen out of fifteen keywords were processed as part of the last batch.

[0083] The user interface **300G** can include a new batch iteration element **315G**. The new batch iteration element **315G** can include a user interactable element for sampling a new batch of keywords. The keyword filter system **100** is configured to receive input through the element **315G** and begin another batch of training in response to the input.

[0084] The user interface **300G** can include stop training and generate list element **320G**. The element **320G** can include a user interactable element for providing input to the keyword filter system **100**, which when received, can cause the keyword filter system **100** to stop training and to generate an output list of keywords with recommendations and explanations that were generated up to the point in which input is received by the element **320G**.

[0085] The user interface **300G** can include a score of remaining keywords element **325G**. The keyword filter system **100** is configured to receive input from the element **325G**, and in response, generate recommendations and/or explanations for the remaining keywords in the input list. The keyword filter system **100** can output an output list of keywords as described herein.

[0086] In some examples, the keyword filter system **100** generates two separate output lists, a list for keywords recommended to remove from the input list, and a list of keywords from the input list that have been recommended by the machine learning model for keeping. The keyword filter system **100** can send an output list for user confirmation to update the current keyword lists corresponding to the user.

[0087] The keyword filter system **100** can automatically update the current keyword lists that were received as input to the AI model, with the output list. In some examples, the keyword filter system **100** automatically updates the current keyword lists with the output list, without additional user confirmation. In some examples, the keyword filter system **100** does not update the current keyword list with the output list, but instead provides the output list for download. The keyword filter system **100** is configured to later receive the output list as input for updating the current keyword list, for example after additional inspection of the output list by a user, or some other additional processing of the keyword list.

[0088] FIG. **4A** is a flow diagram of an example process **400A** for filtering a keyword list using a trained AI model, according to aspects of the disclosure. The example process **400A** can be performed on a system of one or more processors in one or more locations, such as the keyword filter system **100** of FIG. **1**.

[0089] The system receives a plurality of keywords, according to block **405**. The keywords can be part of one or more lists of keywords from a content provider. The keywords can be negative keywords or positive keywords. The keywords can be received through one or more user interfaces, or, in some examples, received from a repository storing keyword lists.

[0090] The system receives digital content associated with a user, according to block **410**. The user

can be a content provider providing digital content associated with keywords received according to block **405**. The digital content can include one or more one or more digital components, e.g., text, audio, video, etc., provided by the content provider or from another source, such as a website, mobile application, or the like. The website can be, for example, a web page authored by the content provider or otherwise associated with the content provider.

[0091] In some examples, the system generates a summary of the received digital content. In other examples, the system receives, as the digital content, a summary for providing as input to the one or more AI models of the keyword filter system. In some examples, the summary can be updated using feedback data, e.g., additions, deletions, and/or modifications of the generated summary. The feedback data can be provided by the content provider or another source.

[0092] The system generates, for each keyword of the plurality of keywords, a recommendation to remove the keyword from the plurality of keywords or to keep the keyword in the plurality of keywords, according to block **415**. The recommendation to remove or keep the keyword can be generated, for example, as metadata or as part of a separate column or field associated with each keyword. The system uses one or more AI models trained as described herein to generate the recommendation.

[0093] The system generates, for each keyword, a respective natural language explanation of the respective recommendation for the keyword, according to block **420**. The explanation can provide an explanation for the recommendation of the system to keep or remove the keyword.

[0094] In some examples, the system generates a recommendation without a respective explanation. In some use cases, generating a recommendation alone is sufficient, and therefore additional bandwidth to communicate the output generated by the system to a user computing device is not needed. In the aggregate, the keyword filter system can reduce data traffic for multiple content providers communicating with the system through multiple computing devices, improving overall operation of the system in use cases in which the content provider does not require an explanation for the recommendations by the system. In some examples, the system is configured to generate explanations in response to a separate request for explanations for keyword recommendations, allowing the explanations to still be accessed on an ad hoc or as-requested basis.

[0095] The system generates an output list of keywords from the plurality of keywords, according to block **430**.

[0096] The system provides the output list of keywords, according to block **440**. The output list can be provided in response to a request for digital content, for example a request received to a digital content delivery system. The digital content delivery system can determine, using the output list and the input including a digital content request, whether digital content associated with the output list should be provided in response to the request. If the digital content delivery system determines that digital content associated with the output list should be provided in response to the request, the system can proceed to transmit the digital content.

[0097] FIG. **4B** is a flow diagram of an example process for training an AI model to filter a keyword list, according to aspects of the disclosure. The example process can be performed on a system of one or more processors in one or more locations, such as the keyword filter system **100** of FIG. **1**.

[0098] The system divides an input keyword list of a plurality of keywords into a plurality of batches, according to block **450**. Each batch can include, for example, ten to twenty keywords, although in some examples a batch can include a single keyword or include more than twenty keywords. The system can perform one or more iterations of the operations according to blocks **455-465**.

[0099] For a batch of keywords, the system generates, using one or more AI models, output including a respective recommendation and a respective explanation for each keyword in the one or more batches, according to block **455**. The one or more AI models receives the batch of keywords, as well as a summary for the content provider associated with the keywords. The one or more AI

models can include a model trained to generate the summary from digital content, as described herein. The one or more AI models can include a single model trained to generate a summary, as well as to generate a recommendation and explanation to keep or remove a keyword from the batch of keywords received. The one or more AI models can include a large generative model, such as a large language model (LLM). The one or more AI models can already be trained on a larger corpus of training data, and then further trained or fine-tuned according to the process **400B**, described herein.

[0100] The system receives feedback data in response to the generated output, according to block **460**. The system can provide, through a user interface, the output recommendations, and explanations for each keyword in a received batch. The feedback data can be received by the system through the user interface.

[0101] The system updates the one or more AI models using the feedback data, according to block **465**. The system can update weight values, bias values, and/or other parameter values representing the one or more AI models, to reflect changes to the models in response to the feedback data. For example, the one or more AI models can be updated using a supervised fine-tuning process, in which the feedback data is used by a training engine performing the fine-tuning to provide updated labels corresponding to the keywords of the received batch. The updated labels can include a ground-truth recommendation and/or explanation of the recommendation, provided as part of the feedback data.

[0102] After one or more iterations of training batches of keywords, the system can receive input, e.g., through the user interface, to stop training and process the remaining keywords. The remaining keywords are processed through the one or more AI models, which have been fine-tuned according to the iterations of training batches of keywords. The system can generate output recommendations and explanations for keywords, which were not part of the original training batches. This process of training the one or more AI models allows the system to generate more accurate recommendations, based on specific context or content delivery strategies informed by feedback data provided by the content provider or another reviewer. The keyword filter system trained according to aspects of the disclosure overall provides more accurate keyword lists, resulting in less wasted bandwidth caused by erroneous digital content delivery. The output lists also may contain far fewer keywords than the original input lists, which allows for more efficient storage of the keyword lists using less memory or data storage, overall.

[0103] Implementations of the present technology include, but are not limited to, the following:

[0104] (1) A method, when performed by one or more processors, causes the one or more processors to perform operations including: receiving, by one or more processors, a plurality of keywords; receiving, by the one or more processors, digital content associated with a user; executing, by the one or more processors, an artificial intelligence (AI) model, wherein executing the AI model comprises: generating, for each keyword of the plurality of keywords, a recommendation to remove the keyword from the plurality of keywords or to keep the keyword in the plurality of keywords, wherein the recommendation is based on the digital content associated with the user, and generating an output list of keywords from the plurality of keywords, wherein each keyword with a respective recommendation to remove the keyword is omitted from the list; and providing as output, by the one or more processors, the output list of keywords. [0105] (2) The method of (1), wherein receiving the digital content associated with the user includes: receiving one or more digital components associated with the user; and generating, as the digital content, a summary of the one or more digital components associated with the user. [0106] (3) The method of (1) or (2), further including updating the summary of the one or more digital components in accordance with feedback data received through a user interface. [0107] (4) The method of any one of (1) through (3), wherein the one or more digital components includes a web page associated with the user. [0108] (5) The method of any one of (1) through (4), further including training the AI model, the training including: dividing the plurality of keywords into a plurality of batches; and for

one or more batches in the plurality of batches: generating output comprising a respective recommendation and a respective natural language explanation for the respective recommendation for each keyword in the one or more batches, receiving feedback data in response to the generated output, and updating the AI model using the feedback data. [0109] (6) The method of (5), wherein the method further includes, after updating the AI model using the feedback data, generating output comprising a respective recommendation and a respective explanation for each keyword of a batch of the plurality of batches that is not of the one or more batches. [0110] (7) The method of (5) or (6), wherein the AI model is a machine learning model or a large language model. [0111] (8) The method of any one of (1) through (7), further including: receiving, by the one or more processors, input from a computing device, the input comprising natural language and associated with a request for digital content; determining, by the one or more processors and based on the output list of keywords, whether the input comprises one or more keywords in the output list of keywords; and in response to determining that the input does not comprise one or more keywords in the output list of keywords, providing digital content in response to the request. [0112] (9) The method of (8), wherein the input is a search query comprising a request for digital content. [0113] (10) A system including memory and one or more processors, the one or more processors configured to perform a method as in any one of (1) through (9). [0114] (11) One or more non-transitory computer-readable storage media storing instructions that are operable, when executed by one or more processors, to cause the one or more processors to perform operations including the method of any one of (1) through (9).

[0115] In some implementations, the techniques disclosed herein enable artificial intelligence to perform digital content summarization and keyword list filtering. Artificial intelligence (AI) is a segment of computer science that focuses on the creation of models that can perform tasks with little to no human intervention. Artificial intelligence systems can utilize, for example, machine learning, natural language processing, and computer vision. Machine learning, and its subsets, such as deep learning, focus on developing models that can infer outputs from data. The outputs can include, for example, predictions and/or classifications. Natural language processing focuses on analyzing and generating human language. Computer vision focuses on analyzing and interpreting images and videos. Artificial intelligence systems can include generative models that generate new content, such as images, videos, text, audio, and/or other content, in response to input prompts and/or based on other information.

[0116] Example machine-learned models include neural networks or other multi-layer non-linear models. Example neural networks include feed forward neural networks, deep neural networks, recurrent neural networks, and convolutional neural networks. Some example machine-learned models can leverage an attention mechanism such as self-attention. For example, some machine-learned models can include multi-headed self-attention models (e.g., transformer models).

[0117] The model(s) can be trained using various training or learning techniques. The training can implement supervised learning, unsupervised learning, reinforcement learning, etc. The training can use techniques such as, for example, backwards propagation of errors. For example, a loss function can be backpropagated through the model(s) to update one or more parameters, e.g., weights or biases, of the model(s) (e.g., based on a gradient of the loss function). Various loss functions can be used such as mean squared error, likelihood loss, cross entropy loss, hinge loss, and/or various other loss functions. Gradient descent techniques can be used to iteratively update the parameters over a number of training iterations. A number of generalization techniques (e.g., weight decays, dropouts) can be used to improve the generalization capability of the models being trained.

[0118] For example, training data can include multiple training examples that can be received as input by the machine learning models. The training examples can be labeled with a desired output for the model when processing the labeled training examples. The training examples can be labeled with noisy labels that guarantee label differential privacy. The noisy labels and the model output

can be evaluated through a loss function to determine an error, which can be back propagated through the machine learning model to update weights for the model.

[0119] The model(s) can be pre-trained before domain-specific alignment. For instance, a model can be pre-trained over a general corpus of training data and fine-tuned on a corpus of training data, e.g., training data including feedback data from a user interface as described herein. A model can be aligned using prompts that are designed to elicit domain-specific outputs, e.g., for keyword filtering according to aspects of the disclosure. Prompts can be designed to include learned prompt values (e.g., soft prompts). The trained model(s) may be validated prior to their use using input data other than the training data and may be further updated or refined during their use based on additional feedback/inputs.

[0120] FIG. 5 is a block diagram illustrating one or more model architectures **505**, such as for deployment in a data center **550** housing a hardware accelerator **510** on which the deployed models will execute for keyword list filtering, according to aspects of the disclosure. The hardware accelerator **510** can be any type of processor, such as a CPU, GPU, FPGA, or ASIC such as a Tensor Processing Unit (TPU). Although only one hardware accelerator **510** is shown as part of the data center **550**, it is understood that multiple hardware accelerators can be implemented across different physical locations, including across different data centers, including the data center **550**.

[0121] An architecture of a model can refer to characteristics defining the model, such as characteristics of layers for the model, how the layers process input, or how the layers interact with one another. For example, the model can be a convolutional neural network that includes a convolution layer that receives input data, followed by a pooling layer, followed by a fully connected layer that generates a result. The architecture of the model can also define types of operations performed within each layer. For example, the architecture of a convolutional neural network may define that rectified linear unit (ReLU) activation functions are used in the fully connected layer of the network. One or more model architectures can be generated that can output results associated with summary generation and keyword list filtering.

[0122] FIG. 6 is a block diagram of an example computing environment **600** for implementing the keyword filter system **100**. The keyword filter system **100** can be implemented on one or more devices having one or more processors in one or more locations, such as in server computing device **615**. User computing device **612** and the server computing device **615** can be communicatively coupled to one or more storage devices **630** over a network **660**. The storage device(s) **630** can be a combination of volatile and non-volatile memory and can be at the same or different physical locations than the computing devices **612**, **615**. For example, the storage device(s) **630** can include any type of non-transitory computer readable medium capable of storing information, such as a hard-drive, solid state drive, tape drive, optical storage, memory card, ROM, RAM, DVD, CD-ROM, write-capable, and read-only memories.

[0123] The server computing device **615** can include one or more processors **613** and memory **614**. The memory **614** can store information accessible by the processor(s) **613**, including instructions **621** that can be executed by the processor(s) **613**. The memory **614** can also include data **623** that can be retrieved, manipulated, or stored by the processor(s) **613**. The memory **614** can be a type of non-transitory computer readable medium capable of storing information accessible by the processor(s) **613**, such as volatile and non-volatile memory. The processor(s) **613** can include one or more central processing units (CPUs), graphic processing units (GPUs), field-programmable gate arrays (FPGAs), and/or application-specific integrated circuits (ASICs), such as tensor processing units (TPUs).

[0124] The instructions **621** can include one or more instructions that when executed by the processor(s) **613**, causes the one or more processors to perform actions defined by the instructions. The instructions **621** can be stored in object code format for direct processing by the processor(s) **613**, or in other formats including interpretable scripts or collections of independent source code modules that are interpreted on demand or compiled in advance. The instructions **621** can include

instructions for implementing the keyword filter system **100** consistent with aspects of this disclosure. The keyword filter system **100** can be executed using the processor(s) **613**, and/or using other processors remotely located from the server computing device **615**.

[0125] The data **623** can be retrieved, stored, or modified by the processor(s) **613** in accordance with the instructions **621**. The data **623** can be stored in computer registers, in a relational or non-relational database as a table having a plurality of different fields and records, or as JSON, YAML, proto, or XML documents. The data **623** can also be formatted in a computer-readable format such as, but not limited to, binary values, ASCII, or Unicode. Moreover, the data **623** can include information sufficient to identify relevant information, such as numbers, descriptive text, proprietary codes, pointers, references to data stored in other memories, including other network locations, or information that is used by a function to calculate relevant data.

[0126] The user computing device **612** can also be configured similar to the server computing device **615**, with one or more processors **616**, memory **617**, instructions **618**, and data **619**. The user computing device **612** can also include a user output **626**, and a user input **624**. The user input **624** can include any appropriate mechanism or technique for receiving input from a user, such as keyboard, mouse, mechanical actuators, soft actuators, touchscreens, microphones, and sensors.

[0127] The server computing device **615** can be configured to transmit data to the user computing device **612**, and the user computing device **612** can be configured to display at least a portion of the received data on a display implemented as part of the user output **626**. The user output **626** can also be used for displaying an interface between the user computing device **612** and the server computing device **615**. The user output **626** can alternatively or additionally include one or more speakers, transducers or other audio outputs, a haptic interface or other tactile feedback that provides non-visual and non-audible information to the platform user of the user computing device **612**.

[0128] Although FIG. 6 illustrates the processors **613**, **616** and the memories **614**, **617** as being within the computing devices **615**, **612**, components described in this specification, including the processors **613**, **616** and the memories **614**, **617** can include multiple processors and memories that can operate in different physical locations and not within the same computing device. For example, some of the instructions **621**, **618** and the data **623**, **619** can be stored on a removable SD card and others within a read-only computer chip. Some or all of the instructions and data can be stored in a location physically remote from, yet still accessible by, the processors **613**, **616**. Similarly, the processors **613**, **616** can include a collection of processors that can perform concurrent and/or sequential operation. The computing devices **615**, **612** can each include one or more internal clocks providing timing information, which can be used for time measurement for operations and programs run by the computing devices **615**, **612**.

[0129] The server computing device **615** can be configured to receive requests to process data from the user computing device **612**. For example, the environment **600** can be part of a computing platform configured to provide a variety of services to users, through various user interfaces and/or APIs exposing the platform services. One or more services can be a machine learning framework or a set of tools for generating neural networks or other machine learning models according to a specified task and training data. The user computing device **612** may receive and transmit data specifying requests for digital content, search queries, or other data.

[0130] The devices **612**, **615** can be capable of direct and indirect communication over the network **660**. The devices **615**, **612** can set up listening sockets that may accept an initiating connection for sending and receiving information. The network **660** itself can include various configurations and protocols including the Internet, World Wide Web, intranets, virtual private networks, wide area networks, local networks, and private networks using communication protocols proprietary to one or more companies. The network **660** can support a variety of short- and long-range connections. The short- and long-range connections may be made over different bandwidths, such as 2.402 GHz to 2.480 GHz (commonly associated with the Bluetooth® standard), 2.4 GHz and 5 GHz

(commonly associated with the Wi-Fi® communication protocol); or with a variety of communication standards, such as the LTE® standard for wireless broadband communication. The network **660**, in addition or alternatively, can also support wired connections between the devices **612**, **615**, including over various types of Ethernet connection.

[0131] The computing environment **600** can also include a data center **550**, including hardware accelerators A-N, for training or executing the AI models of the keyword filter system **100**.

Although a single server computing device **615**, user computing device **612**, and data center **550** are shown in FIG. **6**, it is understood that the aspects of the disclosure can be implemented according to a variety of different configurations and quantities of computing devices, including in paradigms for sequential or parallel processing, or over a distributed network of multiple devices. In some implementations, aspects of the disclosure can be performed on a single device, and any combination thereof.

[0132] Aspects of this disclosure can be implemented in digital electronic circuitry, in tangibly-embodied computer software or firmware, and/or in computer hardware, such as the structure disclosed herein, their structural equivalents, or combinations thereof. Aspects of this disclosure can further be implemented as one or more computer programs, such as one or more modules of computer program instructions encoded on one or more tangible non-transitory computer storage media for execution by, or to control the operation of, one or more data processing apparatus. The computer storage medium can be a machine-readable storage device, a machine-readable storage substrate, a random or serial access memory device, or combinations thereof. The computer program instructions can be encoded on an artificially generated propagated signal, such as a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus.

[0133] The term “configured” is used herein in connection with systems and computer program components. For a system of one or more computers to be configured to perform particular operations or actions means that the system has installed software, firmware, hardware, or a combination thereof that cause the system to perform the operations or actions. For one or more computer programs to be configured to perform operations or actions means that the one or more programs include instructions that, when executed by one or more data processing apparatus, cause the apparatus to perform the operations or actions.

[0134] The term “data processing apparatus” refers to data processing hardware and encompasses various apparatus, devices, and machines for processing data, including programmable processors, a computer, or combinations thereof. The data processing apparatus can include special purpose logic circuitry, such as a field programmable gate array (FPGA) or an application specific integrated circuit (ASIC), such as a Tensor Processing Unit (TPU). The data processing apparatus can include code that creates an execution environment for computer programs, such as code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or combinations thereof.

[0135] The data processing apparatus can include special-purpose hardware accelerator units for implementing machine learning models to process common and compute-intensive parts of machine learning training or production, such as inference or workloads. Machine learning models can be implemented and deployed using one or more machine learning frameworks, such as static or dynamic computational graph frameworks.

[0136] The term “computer program” refers to a program, software, a software application, an app, a module, a software module, a script, or code. The computer program can be written in any form of programming language, including compiled, interpreted, declarative, or procedural languages, or combinations thereof. The computer program can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. The computer program can correspond to a file in a file system and can be

stored in a portion of a file that holds other programs or data, such as one or more scripts stored in a markup language document, in a single file dedicated to the program in question, or in multiple coordinated files, such as files that store one or more modules, sub programs, or portions of code. The computer program can be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a data communication network.

[0137] The term “database” refers to any collection of data. The data can be unstructured or structured in any manner. The data can be stored on one or more storage devices in one or more locations. For example, an index database can include multiple collections of data, each of which may be organized and accessed differently.

[0138] The term “engine” refers to a software-based system, subsystem, or process that is programmed to perform one or more specific functions. The engine can be implemented as one or more software modules or components or can be installed on one or more computers in one or more locations. A particular engine can have one or more computers dedicated thereto, or multiple engines can be installed and running on the same computer or computers.

[0139] The processes and logic flows described herein can be performed by one or more computers executing one or more computer programs to perform functions by operating on input data and generating output data. The processes and logic flows can also be performed by special purpose logic circuitry, or by a combination of special purpose logic circuitry and one or more computers.

[0140] A computer or special purpose logic circuitry executing the one or more computer programs can include a central processing unit, including general or special purpose microprocessors, for performing or executing instructions and one or more memory devices for storing the instructions and data. The central processing unit can receive instructions and data from the one or more memory devices, such as read only memory, random access memory, or combinations thereof, and can perform or execute the instructions. The computer or special purpose logic circuitry can also include, or be operatively coupled to, one or more storage devices for storing data, such as magnetic, magneto optical disks, or optical disks, for receiving data from or transferring data to. The computer or special purpose logic circuitry can be embedded in another device, such as a mobile phone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS), or a portable storage device, e.g., a universal serial bus (USB) flash drive, as examples.

[0141] Computer readable media suitable for storing the one or more computer programs can include any form of volatile or non-volatile memory, media, or memory devices. Examples include semiconductor memory devices, e.g., EPROM, EEPROM, or flash memory devices, magnetic disks, e.g., internal hard disks or removable disks, magneto optical disks, CD-ROM disks, DVD-ROM disks, or combinations thereof.

[0142] Aspects of the disclosure can be implemented in a computing system that includes a back-end component, e.g., as a data server, a middleware component, e.g., an application server, or a front-end component, e.g., a client computer having a graphical user interface, a web browser, or an app, or any combination thereof. The components of the system can be interconnected by any form or medium of digital data communication, such as a communication network. Examples of communication networks include a local area network (LAN) and a wide area network (WAN), e.g., the Internet.

[0143] The computing system can include clients and servers. A client and server can be remote from each other and interact through a communication network. The relationship of client and server arises by virtue of the computer programs running on the respective computers and having a client-server relationship to each other. For example, a server can transmit data, e.g., an HTML page, to a client device, e.g., for purposes of displaying data to and receiving user input from a user interacting with the client device. Data generated at the client device, e.g., a result of the user interaction, can be received at the server from the client device.

[0144] Unless otherwise stated, the foregoing alternative examples are not mutually exclusive, but may be implemented in various combinations to achieve unique advantages. As these and other variations and combinations of the features discussed above can be utilized without departing from the subject matter defined by the claims, the foregoing description of the embodiments should be taken by way of illustration rather than by way of limitation of the subject matter defined by the claims. In addition, the provision of the examples described herein, as well as clauses phrased as “such as,” “including” and the like, should not be interpreted as limiting the subject matter of the claims to the specific examples; rather, the examples are intended to illustrate only one of many possible embodiments. Further, the same reference numbers in different drawings can identify the same or similar elements.

Claims

1. A method, comprising: receiving, by one or more processors, a plurality of keywords associated with digital content of a user, the plurality of keywords being used to identify the digital content responsive to a search query or request for content; generating, by the one or more processors, a text summary of one or more digital components of the digital content associated with the user; executing, by the one or more processors, an artificial intelligence (AI) model, wherein executing the AI model comprises: receiving, as input, the plurality of keywords and the text summary of the one or more digital components; generating, for each keyword of the plurality of keywords, a recommendation to remove the keyword from the plurality of keywords or to keep the keyword in the plurality of keywords, wherein the recommendation is based on the text summary of the one or more digital components, and generating an output list of keywords from the plurality of keywords, wherein each keyword with a respective recommendation to remove the keyword is omitted from the output list; and providing as output, by the one or more processors, the output list of keywords; and; updating the plurality of keywords associated with the digital content of the user with the output list of keywords.
2. (canceled)
3. The method of claim 1, wherein the method further comprises updating the text summary of the one or more digital components in accordance with feedback data received through a user interface.
4. The method of claim 1, wherein the one or more digital components comprises a web page associated with the user.
5. The method of claim 1, further comprising training the AI model, the training comprising: dividing the plurality of keywords into a plurality of batches; and for one or more batches in the plurality of batches: generating output comprising a respective recommendation and a respective natural language explanation for the respective recommendation for each keyword in the one or more batches, receiving feedback data in response to the generated output, and updating the AI model using the feedback data.
6. The method of claim 5, wherein the method further comprises, after updating the AI model using the feedback data, generating output comprising a respective recommendation and a respective explanation for each keyword of a batch of the plurality of batches that is not of the one or more batches.
7. The method of claim 5, wherein the AI model is a machine learning model or a large language model.
8. The method of claim 21, further comprising: receiving, by the one or more processors, input from a computing device, the input comprising natural language; determining, by the one or more processors, whether the input comprises one or more positive keywords or one or more negative keywords in the output list of keywords; and providing at least one of the digital components when the input comprises at least one of the positive keywords and none of the negative keywords in the output list of keywords.

9. The method of claim 8, wherein the input is a search query.

10. A system, comprising: memory; and one or more processors configured to: receive a plurality of keywords associated with digital content of a user, the plurality of keywords being used to identify the digital content responsive to a search query or request for content; generate a text summary of one or more digital components of the digital content associated with the user; execute an artificial intelligence (AI) model, wherein to execute the AI model, the one or more processors are configured to: receive, as input, the plurality of keywords and the text summary of the one or more digital components generate, for each keyword of the plurality of keywords, a recommendation to remove the keyword from the plurality of keywords or to keep the keyword in the plurality of keywords, wherein the recommendation is based on the text summary of the one or more digital components, and generate an output list of keywords from the plurality of keywords, wherein each keyword with a respective recommendation to remove the keyword is omitted from the output list; and provide as output, by the one or more processors, the output list of keywords; and update the plurality of keywords associated with the digital content of the user with the output list of keywords.

11. (canceled)

12. The system of claim 10, wherein the one or more processors are further configured to update the text summary of the one or more digital components in accordance with feedback data received through a user interface.

13. The system of claim 10, wherein the one or more digital components comprises a web page associated with the user.

14. The system of claim 10, wherein the one or more processors are further configured to train the AI model, wherein in training the AI model, the one or more processors are configured to: divide the plurality of keywords into a plurality of batches; and for one or more batches in the plurality of batches: generate output comprising a respective recommendation and a respective natural language explanation for the respective recommendation, for each keyword in the one or more batches, receive feedback data in response to the generated output, and update the AI model using the feedback data.

15. The system of claim 14, wherein after updating the AI model using the feedback data, the one or more processors are configured to generate output comprising a respective recommendation and a respective explanation for each keyword of a batch of the plurality of batches that is not of the one or more batches.

16. The system of claim 14, wherein the AI model is a machine learning model or a large language model.

17. The system of claim 22, wherein the one or more processors are further configured to: receive input from a computing device, the input comprising natural language; determine whether the input comprises one or more positive keywords or one or more negative keywords in the output list of keywords; and provide at least one of the digital components when the input comprises at least one of the positive keywords and none of the negative keywords in the output list of keywords.

18. The system of claim 17, wherein the input is a search query.

19. One or more non-transitory computer-readable storage media, storing instructions that are operable, when executed by one or more processors, to cause the one or more processors to perform operations comprising: receiving a plurality of keywords associated with digital content of a user, the plurality of keywords being used to identify the digital content responsive to a search query or request for content; generating a text summary of one or more digital components of the digital content; executing an artificial intelligence (AI) model, wherein executing the AI model comprises: receiving, as input, the plurality of keywords and the text summary of the one or more digital components; generating, for each keyword of the plurality of keywords, a recommendation to remove the keyword from the plurality of keywords or to keep the keyword in the plurality of keywords, wherein the recommendation is based on the text summary of the one or more digital

components, and generating an output list of keywords from the plurality of keywords, wherein each keyword with a respective recommendation to remove the keyword is omitted from the output list; and providing, as output, the output list of keywords; and updating the plurality of keywords associated with the digital content of the user with the output list of keywords.

20. The computer-readable storage media of claim 19, further comprising training the AI model, the training comprising: dividing the plurality of keywords into a plurality of batches; and for one or more batches in the plurality of batches: generating output comprising a respective recommendation and a respective natural language explanation for the respective recommendation, for each keyword in the one or more batches, receiving feedback data in response to the generated output, and updating the AI model using the feedback data.

21. The method of claim 1, wherein the output list of keywords includes positive keywords and negative keywords.

22. The system of claim 10, wherein the output list of keywords includes positive keywords and negative keywords.
